



Pathology of Infectious Diseases

Lec. 12

Dr.Zainab Hashim

Infectious diseases

- A diseases caused by an infectious agents like viruses , bacteria and fungi . Infectious diseases are an important health problem in the worldwide despite the availability of effective vaccines and antibiotics for many types of infections.



- **Commensals** :are microorganisms that live and share the host's food without causing harm such as :
 1. Bacteria living on the skin .
 2. Vitamin K. producing intestinal bacterial flora, which are normal inhabitants.
- **Pathogens:** are microorganisms that injure the host. They include; viruses, bacteria, fungi, protozoa.
- **Pathogenicity:** is the capacity of a particular microorganism to cause disease.
- **Virulence** : is the degree of pathogenicity.

- **Infection** : Is the presence of microorganism in a part of the body where it is normally absent & where if allowed to multiply, it stimulates a host response & cause infectious disease. The outcome depends on balance between the microorganism's aggressive mechanisms & the host defenses.
- **Invasion**: Is the first step of causing an infection, it depends on the penetration of the hosts' tissues by micro-organisms.



Categories of Infectious Agents

1. Viruses
2. Bacteria
3. Fungi
4. Protozoa
5. Helminths



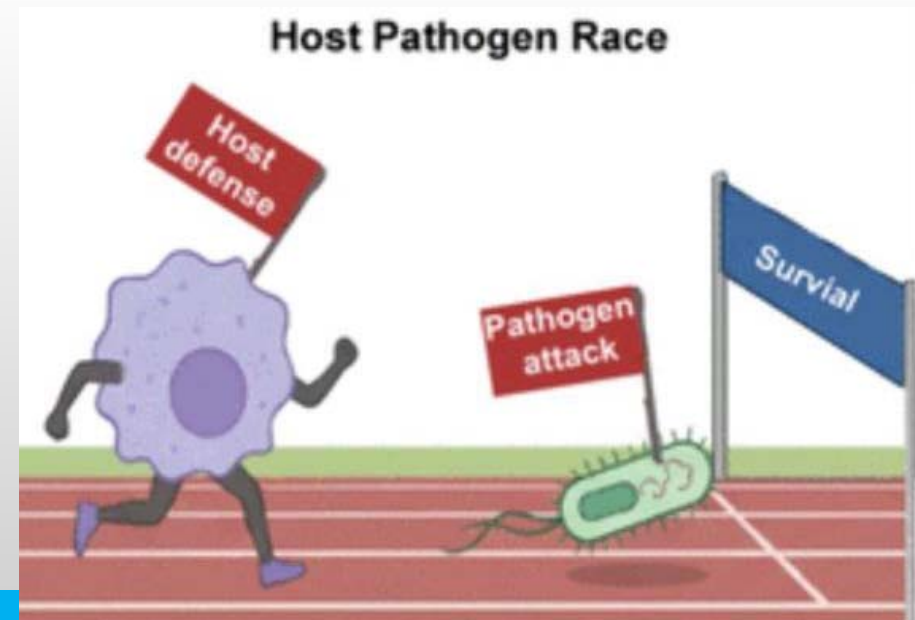
Host defenses against infection

- 1) **Skin:** tough keratinized barrier, low pH, fatty acids, normal microbiota .
- 2) **Respiratory system:** alveolar macrophages and mucociliary clearance by bronchial epithelium, IgA .
- 3) **Gastrointestinal system:** acidic gastric pH, viscous mucus, pancreatic enzymes and bile, defensins, IgA, and normal flora .
- 4) **Urogenital tract:** repeated flushing and acidic environment created by commensal vaginal flora.

Immune evasion by microbes

The microorganism escapes the humoral & cellular immune responses of the host by

- 1-Remaining inaccessible (e.g. residing within host's cells)
- 2-Cleaving antibodies, resisting complement-mediated lysis or surviving in phagocytic cells
- 3-Varying or shedding antigens
- 4-Causing immunosuppression



How microorganisms cause disease ?

- Infectious agents establish infection and damage tissues by any of three mechanisms:
 - 1) They can contact or enter host cells and directly cause death of infected cells.
 - 2) They can release toxins that kill cells at a distance, release enzymes that degrade tissue components, or damage blood vessels and cause ischemic necrosis.
 - 3) They can induce host immune responses that, although directed against the invader, cause additional tissue damage. Thus, the defensive responses of the host can be a mixed blessing, helping to overcome the infection but also contributing to tissue damage.

Patterns of host responses to microbes

In normal (immunocompetent) individuals, the patterns of host responses are fairly stereotypical for different classes of microbes; these response patterns can be used to infer possible causal organisms.

1. Neutrophil-rich acute suppurative inflammation is typical of infections with many bacteria (“pyogenic” bacteria) and some fungi.
2. Mononuclear cell infiltrates are common in many chronic infections and some acute viral infections.

3. Granulomatous inflammation is the hallmark of infection with *M. tuberculosis* and certain fungi.
4. Cytopathic and proliferative lesions are caused by some viruses.
5. Necrosis results from tissue-damaging toxins produced by microbes such as *C. perfringens*.
6. Chronic inflammation and scarring represent the final common pathway of many infections.

Thank You

